

Renewable Energy — Module 5 Study Guide: Environmental & Social Aspects of Sustainability

Course Macro-Framing: The Human Dimension of Sustainability

As humanity navigates planetary boundaries and resource limits, sustainability is not merely an engineering challenge—it is fundamentally a social and ethical framework. Shifting from extractive consumerism to a sustainable future requires balancing public health, clean air protocols, social order, sharing economies, population stability, and biodiversity conservation:

EXTRACTIVE SOCIAL MODEL		SUSTAINABLE SOCIAL FABRIC	
• High industrial pollution & disease	—▶	• Preventive healthcare & clean air	
• Unregulated resource exploitation	—▶	• Sharing economies & social capital	
• Exponential population overshoot	—▶	• Demographic stabilization & equity	
• Habitat fragmentation & coextinction		• Biodiversity conservation & reserves	

- The Health & Environmental Nexus:** Public health is deeply intertwined with environmental conditions. Preventative healthcare and low-carbon strategies, such as afforestation, form the bedrock of sustainable communities.
- Social Capital & Sharing Models:** Moving away from hyper-consumption requires rethinking economic structures through sharing economies, peer-to-peer networks, time banking, and understanding consumption patterns.
- Demographic and Ecological Stability:** Managing population growth, measuring ecological footprints, and protecting biodiversity corridors prevents cascading ecosystem collapse and coextinction.

Lesson 1: Health Care & Lesson 2: Clean Air

Level 1: The Basics (Foundational Concepts)

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Mastery Checklist: Define preventive healthcare, pandemics, afforestation, low-carbon economies, and analyze child survival statistics.

How do healthcare systems and clean air protocols protect global populations from systemic crises?

- **Preventive Health Care:** Utilizing the most successful and cost-effective medical care to ensure the healthiest possible population, stopping illnesses before expensive reactive treatment is required.
- **Pandemic Definition:** An epidemic of infectious disease that spreads across a large geographical region (affecting multiple continents or worldwide).
- **Afforestation:** The process of planting trees in a region where trees did not exist before, creating new carbon sinks to absorb atmospheric carbon dioxide.
- **Low-Carbon Economy:** An economy that releases a minimal amount of greenhouse gas emissions into the atmosphere, transitioning away from fossil fuel combustion.
- **Global Child Survival Baseline:** In the year 2000, approximately **6.3 million to 9.6 million** children under the age of five died worldwide due to preventable health and nutrition crises (with baseline exam models pointing specifically to **9.6 million**). By 2015, global child survival targets under the UN Millennium Development Goals aimed to lift survival rates to **92%**.

Preventive Healthcare & Pandemics —► Low-Carbon Air Protocols —► Afforestation & Carbon Sinks

MIND-BLOWING FACT: In the year 2000, approximately 9.6 million children under five died globally from preventable causes. Achieving the UN MDG survival target of 92% saved millions of young lives worldwide!

EXAM TRAP ALERT: Remember the exact child mortality statistic required on the exam: **9.6 million** children died in the year 2000. Do not confuse afforestation (planting trees where none grew before) with reforestation (replanting cleared forests). Note that Q3 (*nuclear energy*) belongs to Module 1 and is excluded from Module 5 scope.

EXAM CHEAT CODE:

- Cost-effective medical care preventing illness = **Preventive health care**
- Infectious disease spanning continents = **Pandemic**
- Planting trees on non-forested land = **Afforestation**
- Economy minimizing greenhouse gases = **Low-carbon economy**

- Year 2000 under-5 child mortality = **9.6 million**

Level 2: Deepening Understanding (Epidemiological & Atmospheric Interventions)

COVER & TEST YOURSELF:

1. What term defines medical care focused on stopping illnesses before they require expensive treatment?

[Answer: Preventive health care.]

2. Approximately how many children under five died worldwide in the year 2000?

[Answer: 9.6 million children.]

Lesson 3: Cultural Ecology & Sustainability & Lesson 4: The Role of Social Order

Level 1: The Basics (Foundational Concepts)

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Mastery Checklist: Define cultural ecology, cultural selectivity, social practices, social order, and analyze river valley civilizations.

How do human cultures, social institutions, and collective practices shape environmental interaction?

- **Cultural Ecology:** The academic field that studies the interactions between a human society and its natural environment.
- **Cultural Selectivity:** Collective social forums that serve a common purpose based on shared community values.
- **Social Practices:** Routine behaviors and activities that societies perform based on collective customs and cultural values.
- **Social Order:** The social structures, institutions, and practices that perpetuate and stabilize a society.
- **Early Civilizations:** The earliest human civilizations developed primarily along fertile river valleys, including the **Tigris, Euphrates, and Nile** rivers (Ontario is a North American lake, not an ancient river valley).
- **Millennium Development Goals (MDGs):** The United Nations' list of eight targets designed to improve the quality of life and health in developing nations.

Cultural Ecology (Human-Environment Interaction)



Cultural Selectivity (Collective Social Forums)



Social Practices & Order (Institutional Stability)

MIND-BLOWING FACT: Ancient civilizations like those along the Tigris and Nile mastered cultural ecology by aligning their agricultural planting cycles directly with seasonal river floods!

EXAM TRAP ALERT: Pay attention to exam definitions!

- **Cultural Selectivity:** Collective social forums that serve a common purpose.
- **Social Practices:** Things societies do based on collective customs and values.
- **Social Order:** Structures and institutions maintaining society.

EXAM CHEAT CODE:

- Society-environment interaction study = **Cultural ecology**
- Collective social forums for common purpose = **Cultural selectivity**
- Behaviors based on customs/values = **Social practices**
- Structures maintaining society = **Social order**
- UN 8-point development plan = **Millennium Development Goals**

Level 2: Deepening Understanding (Sociological Structures)

COVER & TEST YOURSELF:

1. What term describes collective social forums formed around a common purpose?
[Answer: Cultural selectivity.]
2. Which river valley was NOT one of the cradles of ancient civilization: Tigris, Euphrates, Ontario, or Nile?
[Answer: Ontario (it is a North American lake).]

Lesson 5: Economic Model of Sustainability & Lesson 6: The Shift Towards a Sharing Economy

Level 1: The Basics (Foundational Concepts)

- **Mastery Checklist:** Define sociological water scarcity, patterns of consumption, delayed gratification, the sharing economy, peer-to-peer marketplaces, product service systems,

time banking, social capital, and U.S. resource consumption ratios.

How are modern economic models and individual behaviors shaping resource distribution and collaborative consumption?

- **Sociological Water Scarcity:** A form of water scarcity where physical water may be present, but access is blocked or restricted due to human, institutional, social, or political barriers (e.g., a school bully blocking a thirsty student from using a water vending machine).
- **Patterns of Consumption:** The recurring qualities, acts, behaviors, and systemic tendencies that characterize the ways societies and individuals use products and resources.
- **Delayed Gratification:** The voluntary act of putting off an immediate reward or pleasure now in exchange for a more valuable, sustainable, or important reward later.
- **Sharing Economy:** A socio-economic system based on collaborative consumption where access to products and services does not rely on individual ownership.
- **Peer-to-Peer (P2P) Marketplace:** A decentralized platform where individuals make direct transactions and sharing arrangements with one another.
- **Product Service System (PSS):** A formal commercial business model where a company retains ownership of a product and sells its utility or access as a service to customers.
- **Time Banking:** An institutional system of exchanging reciprocal service hours (such as babysitting or house-sitting) instead of traditional monetary currency.
- **Social Capital:** Networks of relationships among people who live and work in a particular society, enabling cooperation through trust and informal peer favors (e.g., lending your car to a friend).
- **U.S. Resource Disparity:** Though the United States has just **5%** of the world's population, it consumes approximately **25%** of global resources, and Americans throw away roughly **35%** of their food.

[Sharing Economy] → Access over Ownership

- └─ Peer-to-Peer (P2P) Marketplaces (Direct individual trades)
- └─ Product Service Systems (Corporate service leasing)
- └─ Time Banking & Social Capital (Reciprocal service hours & trust)

MIND-BLOWING FACT: The U.S. accounts for only 5% of the global population yet consumes 25% of world resources and discards 35% of its food supply—highlighting the urgent need for sharing and circular economies!

EXAM TRAP ALERT: Know the distinction between formal and informal sharing models! An informal favor between friends (like lending a car) represents **Social Capital**, whereas a commercial service model is a **Product Service System**.

EXAM CHEAT CODE:

- Water access blocked by institutional/human barriers = **Sociological water scarcity**
- Ways societies use products and resources = **Patterns of consumption**
- Putting off immediate rewards for future gain = **Delayed gratification**
- Access over ownership system = **Sharing economy**
- Direct individual transactions = **Peer-to-peer marketplace**
- Exchanging service hours instead of cash = **Time banking**
- Informal peer favors / trust = **Social capital**
- U.S. population share = **5%** | Resource consumption = **25%** | Food waste = **35%**

Level 2: Deepening Understanding (Collaborative Consumption)

COVER & TEST YOURSELF:

1. What economic model allows people to access products and services without purchasing them individually?

[Answer: The sharing economy.]

2. What percentage of global resources does the United States consume with 5% of the world's population?

[Answer: Approximately 25%.]

Lesson 7: Sustainable Populations & Lesson 8: Stabilizing Populations

Level 1: The Basics (Foundational Concepts)

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Mastery Checklist: Define Malthusian theory, the Global Footprint Network, fertility transition, Singapore's demographic trends, China's policy adjustments, and globalization.

How do population growth curves, ecological footprint metrics, fertility rates, and government policies impact global sustainability?

- **Global Footprint Network:** The international research organization and network that measures humanity's ecological footprint and global biocapacity, calculating how fast human consumption outpaces Earth's regeneration.
- **Malthusian Theory:** Thomas Malthus's population theory stating that human population increases exponentially while agricultural food production increases linearly, leading inevitably to resource crises.

- **Fertility Transition:** The demographic shift where birth rates decline as nations industrialize, urbanize, and women gain education and career opportunities.
- **Singapore's Demographic Trend:** Singapore experienced a fertility transition where birth rates fell significantly below replacement level due to delayed marriage ages and urban economic pressures.
- **China's Population Policy:** China amended its strict 1979 one-child policy in recent years to rebalance its rapidly aging population and workforce age ratios.
- **Globalization:** The increasing global interconnectedness between countries on an economic, social, technological, and cultural level.
- **Emerging Countries:** Nations undergoing rapid industrialization and economic transition between developing status and fully industrialized economies.

Global Footprint Network (Biocapacity & Footprint Metrics)



Malthusian Exponential Population Growth vs. Linear Food Supply



Demographic Fertility Transition & Policy Rebalancing (Singapore & China)

MIND-BLOWING FACT: The Global Footprint Network calculates that humanity currently uses ecological resources 1.7 times faster than Earth can renew them!

EXAM TRAP ALERT: Remember Malthus's core premise: population grows **exponentially** while food grows **linear**. Do not invert these growth curves on the exam!

EXAM CHEAT CODE:

- Organization measuring biocapacity and ecological footprint = **Global Footprint Network**
- Exponential population vs linear food = **Malthusian theory**
- Country where fertility fell below replacement from delayed marriage = **Singapore**
- Reason China amended its 1979 policy = **To balance the population age ratio**
- Worldwide interconnectedness = **Globalization**

Level 2: Deepening Understanding (Demographic Shifts)

COVER & TEST YOURSELF:

1. According to Malthusian theory, in what manner does human population increase relative to food supply?

[Answer: Exponentially (while food increases linearly).]

2. What network measures the biologically productive land and water area needed to produce resources and absorb waste?

[Answer: The Global Footprint Network.]

Lesson 9: Biodiversity Conservation & Lesson 10: Species Protection

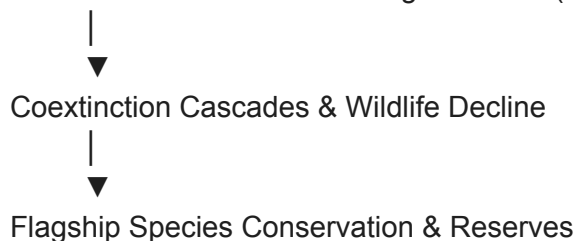
Level 1: The Basics (Foundational Concepts)

- **Mastery Checklist:** Define invasive/invading species, habitat fragmentation, flagship species, coextinction, and IUCN conservation frameworks.

Why is protecting biological diversity and managing non-indigenous species essential for maintaining planetary resilience?

- **Invasive Species (Invading Species):** Any non-indigenous (foreign) living organism that is introduced into a new ecosystem, where it proliferates, outcompetes native species, and disrupts ecological balance.
- **Habitat Fragmentation:** The breaking apart of continuous natural ecosystems into isolated, disconnected patches caused by human infrastructure, agriculture, and urban development.
- **Flagship Species:** Popular, charismatic animals (such as pandas, tigers, or whales) selected as symbols to rally public support and funding for broad environmental conservation causes.
- **Coextinction:** The cascading disappearance of interdependent species that rely on one another for survival (e.g., a specialized flowering plant going extinct when its sole pollinating insect disappears).
- **IUCN Red List:** The world's most comprehensive inventory documenting the global conservation status of biological species.

Invasive Species Introduction & Habitat Fragmentation (Isolated Ecosystem Patches)



MIND-BLOWING FACT: Habitat fragmentation creates ecological "islands," cutting wildlife off from migration corridors and triggering coextinction chains among interdependent species!

EXAM TRAP ALERT: Do not confuse an *invasive species* with a native indicator species. Also, do not confuse a *flagship species* with an indicator species. Flagship species are chosen for their public appeal to drive conservation funding.

EXAM CHEAT CODE:

- Non-indigenous organism disrupting new ecosystem = **Invasive species (Invading species)**
- Breaking ecosystems into isolated patches = **Habitat fragmentation**
- Charismatic symbols for conservation causes = **Flagship species**
- Interdependent species disappearing together = **Coextinction**

Level 2: Deepening Understanding (Ecosystem Resilience)

COVER & TEST YOURSELF:

1. What term describes any non-indigenous organism introduced into an ecosystem that disrupts native life?

[Answer: Invasive species (Invading species).]

2. What term describes the separation of wildlife habitats into isolated patches caused by human development?

[Answer: Habitat fragmentation.]

Comprehensive Module 5 Review & Exam Quick-Reference

| **Concept / Question Target** | **Key Term / Exact Value** | **Specific Exam Context & Contextual Anchor** |

| **Preventative Medical Care** | Preventive health care | Cost-effective medical care ensuring population health. |

| **Multi-Continent Disease** | Pandemic | Infectious disease spreading across multiple continents. |

| **Tree Planting on Bare Land** | Afforestation | Establishing forest cover where none existed before. |

| **Low Greenhouse Gas Economy** | Low-carbon economy | Economy minimizing greenhouse gas emissions. |

| **Year 2000 Under-5 Mortality** | 9.6 million | UN baseline figure for child deaths in 2000. |

| **Society-Environment Study** | Cultural ecology | Field studying human-environment interactions. |

| **Collective Social Forums** | Cultural selectivity | Forums formed around a common purpose. |

| **Customs-Based Behaviors** | Social practices | Routine behaviors based on collective customs and values. |

| **Society Stabilizers** | Social order | Institutions and practices maintaining society. |

| **Water Access Blocked** | Sociological water scarcity | Water is present, but human/institutional barriers block access. |

| **Societal Resource Use Ways** | Patterns of consumption | Qualities, acts, and tendencies characterizing resource use. |

| **Future Reward Postponement** | Delayed gratification | Putting off immediate pleasure for a more important future goal. |

| **Access Over Ownership** | Sharing economy | Collaborative consumption system without ownership. |

| **Direct Individual Trades** | Peer-to-peer marketplace | Direct transactions made between individuals. |

| **Reciprocal Service Hours** | Time banking | Exchanging service hours instead of currency. |

| **Informal Peer Trust** | Social capital | Networks of relationships enabling informal favors. |

| **U.S. Population Share** | 5% | American share of total global population. |

| **U.S. Resource Consumption** | 25% | American share of global resource consumption. |

| **U.S. Food Waste Share** | 35% | Percentage of food thrown away in America. |

| **Biocapacity Measuring Network** | Global Footprint Network | International organization measuring ecological footprints. |

| **Population Growth Curve** | Malthusian theory | Exponential population vs. linear food supply. |

| **Sub-Replacement Fertility** | Singapore | Country where fertility fell due to delayed marriage. |

| **Policy Rebalance Nation** | China | Country amending its 1979 policy to rebalance age ratios. |

| **Worldwide Interconnectedness** | Globalization | Economic, social, and cultural global integration. |

| **Foreign Species Disruption** | Invasive species (Invading species) | Non-indigenous organism

disrupting a new ecosystem. |

| **Ecosystem Splitting** | Habitat fragmentation | Breaking continuous habitats into isolated patches. |

| **Charismatic Symbols** | Flagship species | Animals chosen to rally conservation support. |

| **Cascading Extinction** | Coextinction | Disappearance of interdependent species together. |

If you want to know more about this, visit these sites

Explore global health statistics, preventative care guidelines, and pandemic data:

- [World Health Organization \(WHO\): Global Health Observatory](#) (Confidence: High based on training data)

Review United Nations population trends, fertility rates, and demographic shifts:

- [United Nations Population Division](#) (Confidence: High based on training data)

Investigate ecological footprints, biocapacity, and resource consumption ratios:

- [Global Footprint Network](#) (Confidence: High based on training data)

Examine biodiversity conservation, endangered species, and habitat fragmentation:

- [IUCN Red List & Conservation Hub](#) (Confidence: High based on training data)

Learn more about sharing economies, collaborative consumption, and time banking:

- [Sharing Economy Research Center](#) (Confidence: High based on training data)

Access environmental science lessons on human populations and ecosystems:

- [Khan Academy: AP Environmental Science & Ecology](#) (Confidence: High based on training data)